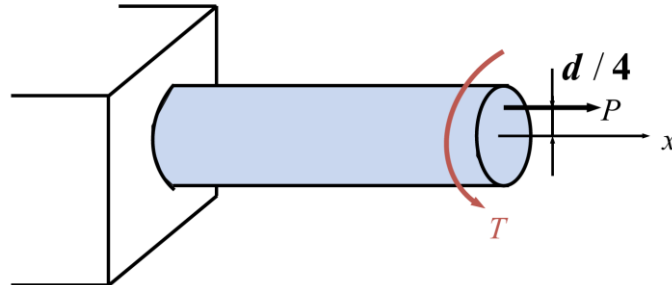


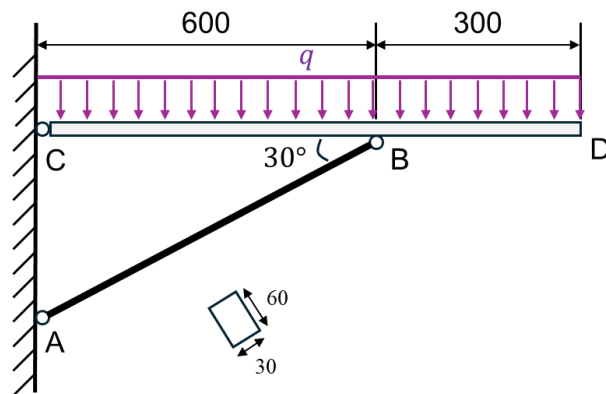
Submission: Please submit the scanned/typed document to Canvas.

1. A solid circular shaft of diameter d is subjected to a torsional moment T and an axial force P . Given that $T = Pd$, identify the location of the critical point (the point with highest principal stress σ_1) and write the corresponding principal stress σ_1



2. A solid circular beam CD has a diameter of $d = 100$ mm, $[\sigma] = 160$ MPa. Bar AB with a rectangular cross section, $E = 200$ GPa, $\sigma_{pl} = 200$ MPa, $\sigma_s = 240$ MPa. The linear relation of inelastic buckling has coefficient $a = 310$ MPa and $b = 1.14$ MPa, The safety factor of stability is $n_{st} = 3$. Ignore the transverse shear and axial force in beam CD. Please determine the allowable load q_{max} .

(Hint: the force acting on the bar AB is along its longitudinal axis)



3. Please determine the critical buckling load of an initially straight prismatic column AB with length L , Young's modulus E , in-plane moment of inertia I . The bottom end, B , is fixed to the ground, while the top end, A , is pinned to a roller that is free to move vertically.

